

Machine Learning Potentials for Simulating Spin-Crossover Materials

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Introduction

Atomic-level simulations have been a vital part of modern materials science and engineering by providing fundamental insights into material properties [1]. Molecular dynamics is a computational method used to study how atoms and molecules move and interact over time. Historically, the methods for these simulations have been divided into two approaches. Ab initio molecular dynamics simulate these systems from quantum first principles providing highly accurate, but computationally expensive results. Alternatively, physics-based potentials use classical theory to inform the nature of atomic interactions resulting in scalable but less accurate simulations. Machine learned interatomic potentials (MLIPs) are a modern approach to atomic-level simulations that aim to bridge the gap in accuracy and scalability of these two methods [2]. MLIPs “learn” the atomic interactions from ab initio data to predict the potential energy surface of material systems. In theory, this leads to results that match the accuracy of ab initio methods without the need to perform expensive calculations [2].

Spin-crossover (SCO) materials are materials that can reversibly switch between high spin (HS) and low spin (LS) electronic states when subject to external stimuli. This switching between states results in changes in their physical properties, leading to an array of possible applications [3]. Computational simulations and predictions play an important role in understanding and developing these materials [4].

The aim of this project is to develop and test MLIP models for SCO materials and use them to predict their behaviour as a function of external parameters such as temperature.

Significance

Since MLIPs combine the accuracy of quantum mechanical methods with the scalability of classical simulations, we should be able to produce accurate, large-scale simulations of SCO materials. This allows us to study larger and more realistic systems with the accuracy of ab initio methods.

The bi-stability and response to external stimuli exhibited by SCO materials make them promising candidates for use in molecular sensors, switches and data storage [3]. This potential provides strong motivation to investigate these materials.

However, the development of SCO materials relies on the experimentation process where candidate materials must be synthesised and then tested. This process requires labour and time with many candidate materials failing to produce SCO behaviour [4]. Therefore, computational simulations are essential to the development of these materials, giving incentive to find efficient and accurate computational methods such as the use of MLIPs.

Methods

To implement models for use in the simulation of SCO materials, we will use MACE and mMACE (magnetic MACE), two of the most promising machine learning frameworks used to develop MLIP models [5, 6]. mMACE is a framework that was recently introduced to overcome the inability of standard MLIPs to simulate complex spin states and magnetic materials [6]. For these frameworks, there are existing foundation models which are models that have been pre-trained on general data sets and are designed to work out of the box. The first 3 weeks will be focused on setting up workflows and running calculations with the foundation models. These calculations will include relaxation, phonons and Gibbs free energy.

The following 3 weeks will be spent fine tuning these models to better fit the specific materials we will be working on. At this point we will also run MD simulations using these fine tuned models.

The final two weeks will be spent finalising the project by returning to the foundation models and using these to test their performance for a larger set of SCO materials.

Week #	Focus of Week
1	Familiarise with MACE and mMACE. The foundation models will be installed and workflows will be set up for running calculations.
2-3	After setting up models and workflows we will run calculations using the foundation models. These calculations will include relaxation, phonons etc
4	This week will be focused on fine tuning the foundation models to run calculations for specific structures
5 - 6	Similar to the calculations run in weeks 2-3, these calculations will also be done for the fine tuned models. At this point we will also run MD simulations
7	We will run the foundation models against larger existing datasets
8	Finalise project report and presentation

Figure 1: Project Timeline

Expected Outcomes

At the end of the project we expect to have a method for running accurate MD simulations of SCO materials using MLIPs, specifically MACE and mMACE models. If shown useful, the methods from this project may aid in further research in these materials by making the study of larger systems at longer timescales more computationally feasible. This would improve the understanding of SCO materials and help guide their development for future technologies.

References

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